Jan. 29 | Lect. 6 | SPIN WAVES in FERROMAGNET

$$\hat{\mathcal{H}} = -\frac{1}{2} J \sum_{r,g} \vec{S}_r \vec{S}_{r+g}$$ let $s = 1/2$ for simplicity.

Ferromagnetic ground state $\uparrow\uparrow\uparrow\uparrow\uparrow$ (product state)

$$\vec{S}_r \vec{S}_{r+g} = S^z_r S^z_{r+g} + \frac{1}{2}\left(S^+_r S^-_{r+g} + S^-_r S^+_{r+g}\right)$$

$$\hat{S}^z |\sigma\rangle = \frac{1}{2}\sigma|\sigma\rangle \quad ; |\sigma\rangle = \begin{pmatrix}1\\0\end{pmatrix} \text{ or } \begin{pmatrix}0\\1\end{pmatrix}$$

$$\hat{S}^+ = \hat{S}^x + i\hat{S}^y = \frac{1}{2}(\sigma^x + i\sigma^y) = \frac{1}{2}\left[\begin{pmatrix}0&1\\1&0\end{pmatrix} + i\begin{pmatrix}0&-i\\i&0\end{pmatrix}\right] = \begin{pmatrix}0&1\\0&0\end{pmatrix}$$

$$\hat{S}^+\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0&1\\0&0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \hat{S}^+|\uparrow\rangle = 0$$

$$\hat{S}^+\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0&1\\0&0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} \Rightarrow \hat{S}^+|\downarrow\rangle = |\uparrow\rangle$$

$$\hat{S}^- = (\hat{S}^+)^+ = \begin{pmatrix}0&0\\1&0\end{pmatrix}$$

$$\hat{S}^-\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0&0\\1&0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \hat{S}^-|\downarrow\rangle = 0$$

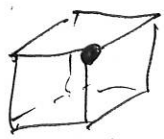
$$\hat{S}^-\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0&0\\1&0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix} \Rightarrow \hat{S}^-|\uparrow\rangle = |\downarrow\rangle$$

$$\Rightarrow \vec{S}_r \vec{S}_{r+g} |\uparrow\rangle_r |\uparrow_{r+g}\rangle = \hat{S}^z_r \hat{S}^z_{r+g} |\uparrow\rangle_r |\uparrow_{r+g}\rangle$$
$$+ \frac{1}{2}\left(\hat{S}^+_r \hat{S}^-_{r+g} + \hat{S}^-_r \hat{S}^+_{r+g}\right)|\uparrow\rangle_r |\uparrow\rangle_{r+g} = \frac{1}{4}|\uparrow\rangle_r |\uparrow\rangle_{r+g}$$

$$\Rightarrow \hat{\mathcal{H}}\left(\prod_r |\uparrow\rangle_r\right) = -\frac{1}{2} J \sum_{rg} \frac{1}{4}\left(\prod_r |\uparrow\rangle_r\right) = -\frac{1}{2} J \left(\frac{1}{4}\right) z N$$

z → # of neighbours

$z = 6$ for simple cubic lattice



$$E_{g.s.} = (-J)\,\frac{1}{4}\cdot N_{bonds}$$

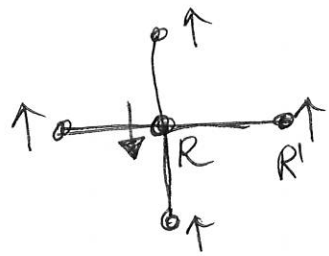
Thus $\prod_r |\uparrow\rangle_r$ is an eigenstate (and, the <u>ground state</u>)

Consider now <u>single defect</u> $|R\rangle = \prod_{r \neq R} |\uparrow\rangle_r |\downarrow\rangle_R$

(Spin down at site R, all others are $\uparrow$)

$$\hat{\mathcal{H}} |R\rangle = -\frac{1}{2} J \sum_{n,g} \left[\hat{S}^z_n \hat{S}^z_{n+g} + \frac{1}{2}\left(\hat{S}^+_n \hat{S}^-_{n+g} + \hat{S}^-_n \hat{S}^+_{n+g}\right)\right] |R\rangle$$

$$\hat{S}^z_R \hat{S}^z_{R'} |\downarrow_R \uparrow_{R'}\rangle =$$



$$= -\frac{1}{4} |\downarrow_R \uparrow_{R'}\rangle$$

But $S^+_R S^-_{R'} |\downarrow_R \uparrow_{R'}\rangle = \underbrace{\hat{S}^+_R |\downarrow\rangle_R}_{|\uparrow\rangle_R} \underbrace{\hat{S}^-_{R'} |\uparrow\rangle_{R'}}_{|\downarrow\rangle_{R'}} = |\uparrow_R \downarrow_{R'}\rangle$

moves $\downarrow$ from $\vec{R}$ to $\vec{R}'$.

while $S^-_R S^+_{R'} |\downarrow_R \uparrow_{R'}\rangle = 0$

$$\Rightarrow \quad S^+_R S^-_{R'} |R\rangle = |R'\rangle \quad !$$

$$\hat{\mathcal{H}} |R\rangle = -\frac{1}{2} J \sum_{ng} \left\{ \left[\frac{1}{4}(1-\delta_{n,R})(1-\delta_{n+g,R}) - \frac{1}{4}\delta_{n,R} - \frac{1}{4}\delta_{n+g,R} \right] |R\rangle \right.$$

$$\left. + \frac{1}{2} \delta_{n,R} |n+g\rangle + \frac{1}{2} \delta_{n+g,R} |n\rangle \right\}$$

$$\cancel{\sum_{n,g} \hat{S}^z_n \hat{S}^z_{n+g} |R\rangle = \sum_{n,g} \frac{1}{4} - \frac{1}{2}}$$

3.

$$S^z_0 (S^z_1 + S^z_2 + S^z_3 + S^z_4)$$

$$\mathcal{H} = -\frac{1}{2} J \underbrace{\sum_{r,g}}_{\text{sum over sites } \vec{r}} \vec{S}_r \cdot \vec{S}_{r+g} = -J \underbrace{\sum_{\langle r r' \rangle}}_{\text{some over bonds } \langle r r' \rangle} \vec{S}_r \cdot \vec{S}_{r'}$$

(every bond appears once)

each bond appears twice, accounted for by factor $\left(\frac{1}{2}\right)$ in front.

$$\Rightarrow \sum_{\langle r r' \rangle} S^z_r S^z_{r'} |R\rangle = \left(\frac{1}{4} \left[\left(\frac{N}{2}\right) - z \right] - \frac{1}{4} z \right) |R\rangle$$

number of bonds in a system with N sites

number of $\downarrow_R \uparrow_{R'}$ bonds = number of nearest-neighbor pairs involving site R.

$$= \frac{N}{8} |R\rangle - \frac{1}{2} z |R\rangle$$

$$\hat{\mathcal{H}}\,|R\rangle = -J\left(\frac{N}{8}-\frac{z}{2}\right)|R\rangle - \frac{1}{4}J\sum_{ng}\left(\delta_{n,R}\,|n+g\rangle + \delta_{n+g,R}\,|n\rangle\right)$$

$$= -\frac{J}{2}\left(\frac{N}{4}-z\right)|R\rangle - \frac{1}{4}J\sum_{g}\left(|R+g\rangle + |R-g\rangle\right)$$

$$= -\frac{J}{2}\left(\frac{N}{4}-z\right)|R\rangle - \frac{1}{2}J\sum_{g}|R+g\rangle$$

$|R\rangle$ is not an eigenstate.

But $|\vec{k}\rangle = \frac{1}{\sqrt{N}}\sum_{R} e^{i\vec{k}\cdot\vec{R}}\,|\vec{R}\rangle$

$$\hat{\mathcal{H}}\,|\vec{k}\rangle = \frac{1}{\sqrt{N}}\sum_{R} e^{i\vec{k}\cdot\vec{R}}\,\hat{\mathcal{H}}\,|\vec{R}\rangle =$$

$$= \frac{1}{\sqrt{N}}\sum_{R} e^{i\vec{k}\cdot\vec{R}}\left[\left(-\frac{JN}{8}+\frac{Jz}{2}\right)|\vec{R}\rangle - \frac{1}{2}J\sum_{g}|\vec{R}+\vec{g}\rangle\right]$$

but $\frac{1}{\sqrt{N}}\sum_{g}\sum_{R} e^{i\vec{k}\cdot\vec{R}}\,|\vec{R}+\vec{g}\rangle = \frac{1}{\sqrt{N}}\sum_{g} e^{-i\vec{k}\cdot\vec{g}}\sum_{R+g} e^{i\vec{k}\cdot(\vec{R}+\vec{g})}\,|\vec{R}+\vec{g}\rangle$

$$= \sum_{g} e^{-i\vec{k}\cdot\vec{g}}\,|\vec{k}\rangle$$

$$\hat{\mathcal{H}}|\vec{k}\rangle = \left(-\frac{JN}{8} + \frac{Jz}{2}\right)|\vec{k}\rangle - \frac{1}{2}J\sum_{g} e^{-i\vec{k}\cdot\vec{g}}|\vec{k}\rangle$$

$$= \underbrace{-\frac{JN}{8}}_{-\frac{JN_{bonds}}{4}}|\vec{k}\rangle + \frac{J}{2}\Big(z - \underbrace{\sum_{g} e^{-i\vec{k}\cdot\vec{g}}}_{2[\cos(k_x a)+\cos(k_y a)+\cos(k_z a)]}\Big)|\vec{k}\rangle$$

$|\vec{k}\rangle$ is eigenstate, with energy (<u>Spin Wave</u>)

$$\varepsilon(\vec{k}) = Jd\left(1 - \frac{1}{d}\sum_{j=1}^{d}\cos(k_j a)\right)$$

(with respect to the ground state energy $-\frac{JN}{8}$)

$$d = \text{dimensionality} = \begin{cases} 3 & \text{cubic lattice} \\ 2 & \text{square} \\ 1 & \text{chain} \end{cases}$$

Small-$\vec{k}$ limit: $\cos(k_j a) \xrightarrow[k\to 0]{} 1 - \frac{1}{2}k_j^2 a^2$

$$\hbar\omega(\vec{k}) = \varepsilon(k) \to Jd\left(1 - \frac{1}{d}\sum_{j=1}^{d}\left(1 - \frac{1}{2}k_j^2 a^2\right)\right)$$

$$= Jd\left(1 - 1 + \frac{1}{2d}(k_x^2 + k_y^2 + k_z^2)a^2\right)$$

$$= \frac{1}{2}Ja^2\vec{k}^2 = Dk^2.$$

* Quadratic dispersion relation for spin waves in ferromagnet.

Number of spin waves

$$N_{SW} = \sum_{\vec{k}} \frac{1}{e^{\frac{\varepsilon(k)}{k_B T}} - 1} = V \int \frac{d\vec{k}}{(2\pi)^3} \frac{1}{e^{\frac{Dk^2}{k_B T}} - 1} =$$

$$= V \int \frac{4\pi}{(2\pi)^3} \int_0^\infty \frac{dk\, k^2}{e^{Dk^2/k_B T} - 1} = \frac{V}{2\pi^2} \frac{1}{2} \left(\frac{k_B T}{D}\right)^{3/2} \int_0^\infty du \frac{u^{1/2}}{e^u - 1}$$

let $u = \frac{Dk^2}{k_B T}$, $k = \sqrt{\frac{u\, k_B T}{D}}$ $\quad dk = \sqrt{\frac{k_B T}{D}} \frac{du}{2\sqrt{u}}$

$\Rightarrow$ per unit volume $\frac{N_{SW}}{V} \sim T^{3/2}$

Each spin wave ~~[illegible]~~ reduces magnetization by $S^z = 1$

$$\Rightarrow \frac{M(0) - M(T)}{M(0)} \approx \alpha T^{3/2}$$

$$M(T) = M(0)\left(1 - \alpha T^{3/2}\right) \quad \underline{\text{Bloch law}}$$

Their energy $E_{SW} = V \int \frac{d\vec{k}}{(2\pi)^3} \frac{\varepsilon(\vec{k})}{e^{\varepsilon(k)/k_B T} - 1} \sim V \int dk \frac{k^2 \cdot Dk^2}{e^{Dk^2/k_B T} - 1}$

$$\sim V \left(\frac{k_B T}{D}\right)^{5/2} D$$

$\Rightarrow$ contribute to specific heat $C_V = \frac{1}{V}\left(\frac{\partial E}{\partial T}\right) \sim T^{3/2}$

(recall, phonons $\hbar\omega \sim ck$, contribute T^3)